A linear Gaussian model defines a joint Gaussian

Linear Gaussian model

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$y = Wz + b + \varepsilon$$
$$\varepsilon \sim \mathcal{N}(0, \Omega)$$

ε is independent of z

Affine Gaussian rules

$$u = Ax + c$$
$$E[u] = AE[x] + c$$
$$\mathrm{Cov}[u] = A\ \mathrm{Cov}[x]A^T$$

Joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix}\right)$$
$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$
$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Murphy, Book 2 §2.3.2.1, Eq. 2.120

- The model plus affine-Gaussian rules determine the full joint distribution
- Once $p(z, y)$ is known, the Gaussian marginal and conditioning rules apply directly

Bayes' rule for Gaussians

Model-specific joint blocks

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix} \right)$$
$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$
$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Gaussian conditioning rule

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Apply: $x_1 = z, x_2 = y$

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$
$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}\left(y - \mu_y\right)$$
$$\Sigma_{z|y} = \Sigma_{zz} - \Sigma_{zy}\Sigma_{yy}^{-1}\Sigma_{yz}$$

Murphy, Book 2 §2.3.2.2, Eq. 2.121

- Apply the Gaussian conditioning rule to the joint blocks
- The mean responds to observed minus predicted measurement
- The covariance subtracts the uncertainty explained by the measurement